MANOJ M JAHGIRDAR

Senior Software Engineer – AI Systems & Backend Engineering

Bengaluru, Karnataka, India • Open to remote / relocation

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SUMMARY

Software engineer with 7+ years building production AI & backend systems. Specialize in agentic AI harness engineering, multi-agent orchestration, RAG pipelines, & LLM-powered platform features. Backed by a strong foundation in backend engineering, Kubernetes, & DevOps. Shipped AI systems that measurably reduced operational effort by 50-80% across financial services, retail, & cloud domains. Patent holder, recognized mentor, contributor to open-source AI tooling & presented as a resource person at IIM Bangalore & other esteem universities. A tech enthusiast, curious in learning technology & applying it to solve real-world problems.

PROFESSIONAL EXPERIENCE

Senior Software Engineer

July 2019 – Present

IBM Software Labs, IBM India Pvt. Ltd. • Bengaluru, Karnataka, India

AI Cloud Developer *(2026 – Present)*

- **Engineered a time-series anomaly detection framework** for VPC observability, using time-series foundation model for matrix imputation and anomaly scoring across the four golden signals. Integrated automated service for onboarding service through AI coding agents. The system caught anomalies in the golden signals pro-actively and notified ahead of an incident **reducing 15 incidents and 2 CIEs so far** and is currently in production.
  Stack: *Python, Prometheus, Grafana, IBM Bob*
- **Extended IBM Cloud IaaS's multi-agent incident-resolution system** for the IaaS CORE team by adding monitoring and alerts spanning across multi-region, enabling the agent to correlate alerts and metrics during triage and find root causes for incidents and CIEs. System **has resolved 70K+ incidents and 500+ CIEs**, adopted by ~300 engineers and consumes <60B tokens/month.
  Stack: *TypeScript, Shell, Python, Claude Code*

AI Engineer *(2024 – 2026)*

- **Owned an end-to-end product development lifecycle of architecting, building and deploying an agentic AI system for financial advisors**, integrating LLM reasoning with live backend systems (portfolio data, CRM, compliance records). The system delivered **~3 hours/day productivity gain per advisor** across multiple enterprise clients.
  Stack: *Python, LangChain Deep Agent, ChromaDB, PostgreSQL, Docling, A2A, MCP, Cursor*
- **Built and scaled a RAG-powered conversational AI assistant** for customer support, resolving thousands of queries monthly over internal policy and product documentation. The assistant achieved a **~65% containment rate** (queries resolved without human escalation).
  Stack: *Python, LangGraph, RAG, FAISS*
- **Designed and deployed a production LLM-powered SRE Agent** integrated into on-call workflows, automating runbook execution and incident triage against production monitoring infrastructure; improved **SLA compliance by ~25%** by reducing MTTR on common incident classes.
  Stack: *Python, LangChain, RAG, Pandas, Shell, Streamlit*

Software Developer *(2019 – 2024)*

- **Built a stateless, auto-scaling backend platform** integrating Twilio WhatsApp webhooks with IBM Watson AI services, serving **~10K DAUs**. Designed for Kubernetes-native horizontal scaling with secure credential management, automated cloud service provisioning, and zero-downtime deployments.
  Stack: *Python, FastAPI, ReactJS, Kubernetes, Db2, MongoDB*
- **Automated cloud infrastructure lifecycle management** for containerized workloads across CI/CD pipelines, eliminating idle resource waste and reducing **annual cloud spend by $159K** while preserving production reliability.
  Stack: *Python, Shell, Go, Tekton, LogDNA, Sysdig*
- **Reduced BCDR operational overhead by ~50%** by automating cross-region secrets backup and synchronization during a Vault-to-IBM-Secrets-Manager migration, eliminating manual rotation steps across all production environments.
  Stack: *Python, Shell, Tekton, LogDNA*

Software Developer Intern *Jan 2019 – Jul 2019*

IBM Software Labs, IBM India Pvt. Ltd. • Bengaluru, Karnataka, India

- Contributed to a cloud-backed personalized retail application adopted by Middle East retail customers. Delivered 4 REST API integrations and an ML-powered product image pipeline serving over **1,000 SKUs**.
  Stack: *Java, Python, TensorFlow, Node.js, MongoDB, Db2, S3, Ionic, Android*

SKILLS

AI & Agentic Systems: Agent Harness Engineering, Loop Engineering, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), Text-to-SQL, Human-in-the-Loop Systems, Prompt Engineering, NLP, Supervised & Unsupervised Learning, Time-series.

Agentic Frameworks & Protocols: LangGraph, LangChain, CrewAI, AutoGen, Google ADK, Model Context Protocol (MCP), Agent-to-Agent Protocol (A2A).

Backend & APIs: Python, Java, Go, JavaScript, FastAPI, ExpressJs, Flask, Django, Spring Boot, RESTful APIs, Microservices

Data & Vector Stores: PostgreSQL, MySQL, MongoDB, Db2, ChromaDB, FAISS, Pinecone, Qdrant, Pandas, NumPy

Cloud, DevOps & MLOps: AWS, GCP, Azure, IBM Cloud, Kubernetes, Docker, Tekton, CI/CD, TDD, SRE Practices

Open-Source: Cloud-native toolkit (4 PRs merged), IBM Granite community (1 PR merged), ContextForge MCP Gateway (1 PR merged)

Recognition: Guest speaker at IIM Bangalore (×2), 1st place in technical paper presentation (×2), IBM Mentor, Cambridge Business English B1 certified.

PATENTS

- **US12392625B2** — System & methodology for persona-based indoor navigation.
- **IPCOM000268675D** — A system and method to predict Customer Lifetime Values based on group persona.

PROJECTS

Browmix ERP — Built a 3-tier backend system on AWS for streamlining manufacturing operations, achieving **3.45% bottom-line improvement**, **8% inventory reduction**, and **15% logistics cost savings** for a corrugated box manufacturing client.

CERTIFICATIONS

Claude Certified Developer Foundations (CCDV-F) – Anthropic

Build Multi-Agent Systems with ADK – Google

Generative AI with Large Language Models – DeepLearning.AI

Machine Learning with python – Stanford University

Azure Fundamentals (AZ-900) – Microsoft

AWARDS

2025 **IBM Growth Award:** Awarded for outstanding performance and business impact.

2022 **IBM People's Choice Award:** Awarded for demonstrating cultural growth and team impact.

2020 **IBM Manager's Choice Award:** Recognized for personal initiative and engineering excellence.

EDUCATION

KLS Gogte Institute of Technology *2015 – 2019*

Bachelor of Engineering in Computer Science and Engineering GPA: 8.89 / 10 • Belagavi, Karnataka, India

INTERESTS

Hardware & Tinkering: Built a Network Attached Storage (NAS) device from repurposed hardware; obstacle-detecting car with Arduino; home automation system with ESP8266; added LLM integration to Alexa.

Sports: Represented my school at district-level badminton tournaments.

Cameras: 1st place, inter-college film making competition.